Data Description

Dataset: Stress analysis from physiological data under pressure: WorkStress3D Dataset

For this study we use physiological signals data and facial expression data from the WorkStress3D dataset. The physiological signal component file incorporates electrodermal activity (EDA), blood volume pressure (BVP), skin temperature, and accelerometer (ACC) data with the corresponding emotion class (0: stress-free, 1: stressful). There are a total of 3 csv files with these data collected over 15, 30, and 60 seconds data window. Similarly, facial expression data consist of pixels from 48 x 48 grayscale images which are categorized as either stress-free (class 0) or stressful (class 1).

Model Implementation

Physiological data:

Features/Input (X): eda, bvp, temp, x, y, z

Label/Output (y): emotion (0:stress-free , 1:stressed)

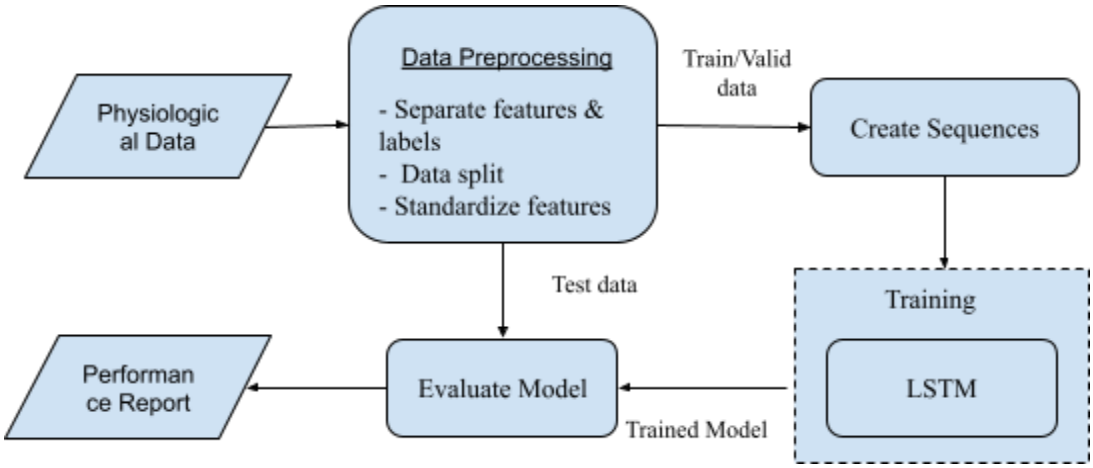


Figure 1. Overall Procedures for implementing LSTM model on physiological data

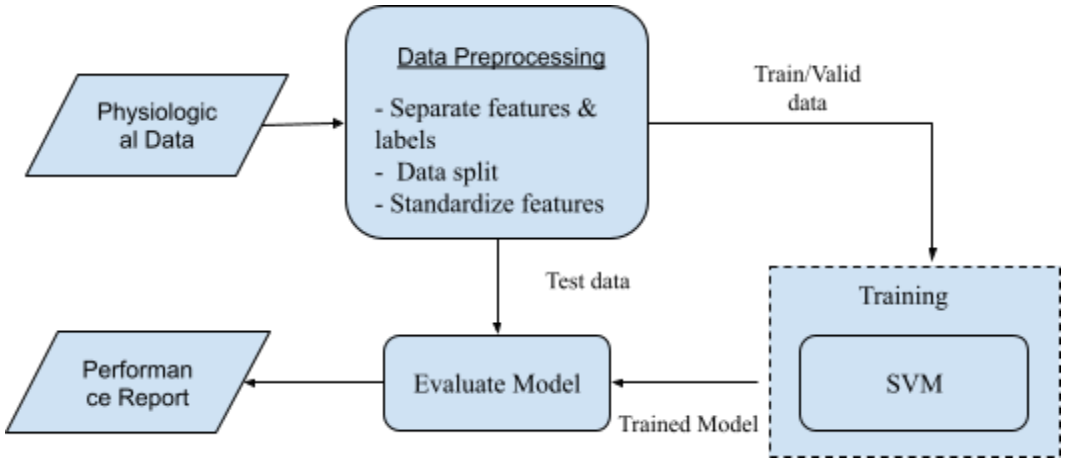


Figure 2. Overall Procedures for implementing SVM model on physiological data

Facial Expression data:

Features/Input (X): 48x48 grayscale pixels
Label/Output (Y): emotions(0: stress-free, 1: stressed)

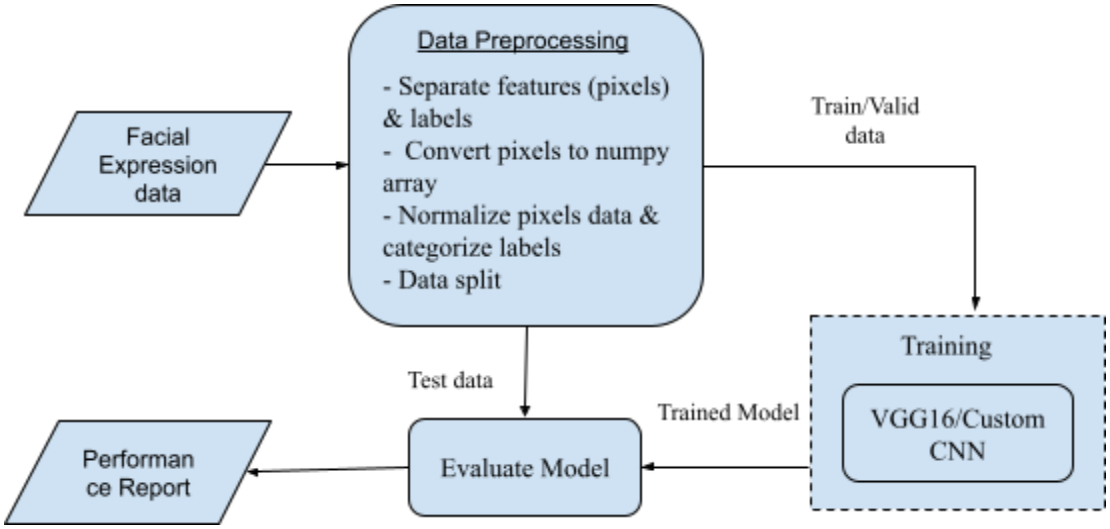


Figure 3. Overall Procedures for implementing CNN model on facial expression data

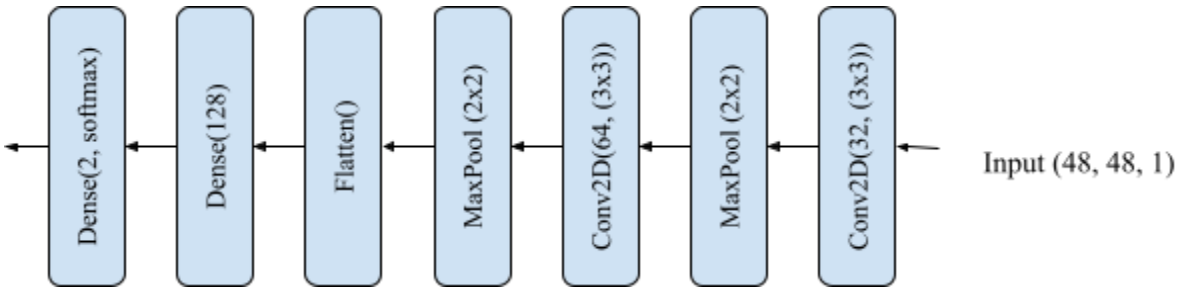


Figure 4. Custom CNN Architecture

Experimentation & Results

Physiological Data

LSTM MODEL:

Data Distribution:

Dataset (Physiological signals data)	num_train	num_validation	num_test	num_items_classes (0,1)
15s data	28062	7016	3898	25704, 13272
30s data	59132	14783	8213	54162, 27966
60s data	120268	30068	16704	110160, 56880

Model Parameters:

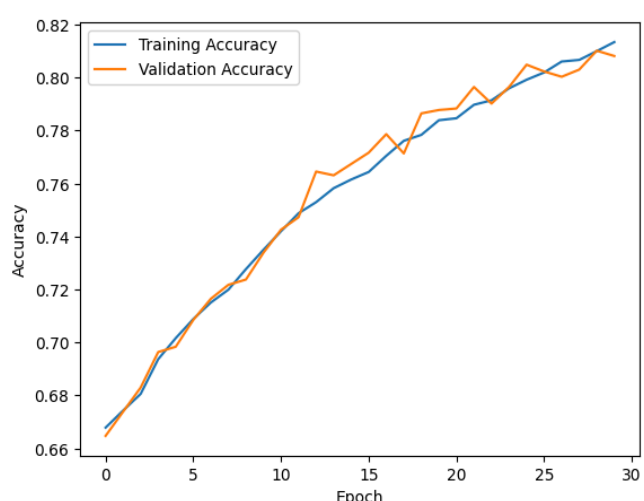
Batch size = 4
Epochs = 30
LSTM Memory Units = 64
Optimizer = Adam
Activation = 'sigmoid'
Loss Function = Binary Cross Entropy
Lr = 0.001

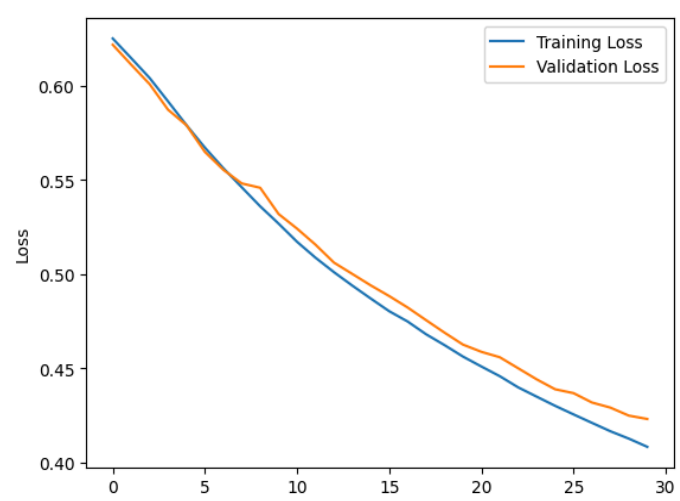
Model Performance:

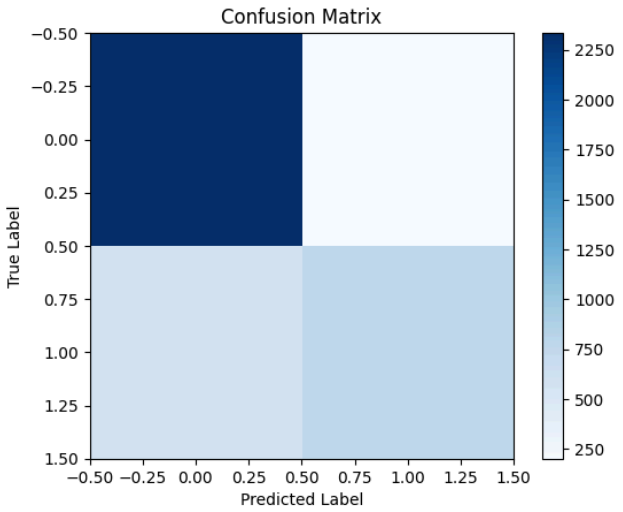
Dataset (Physiological signals)	Model	Train acc	Train loss	Val acc	Val loss	Test acc	precision	recall	f1
15s data	LSTM	0.81	0.40	0.80	0.42	0.80	0.79	0.80	0.79
30s data	LSTM	0.8596	0.3458	0.8583	0.3553	0.85	0.85	0.82	0.83
60s data	LSTM	0.8633	0.3368	0.8615	0.3437	0.86	0.85	0.83	0.84

Graph & Confusion Matrix:

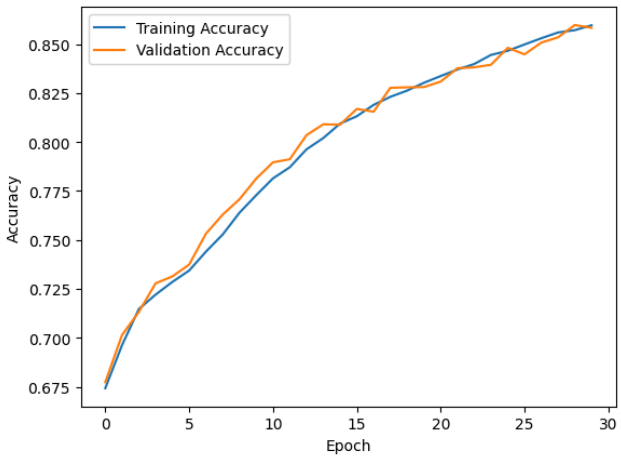
LSTM_15s:

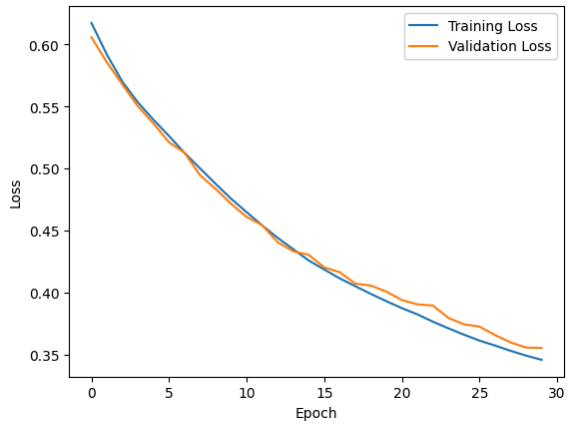


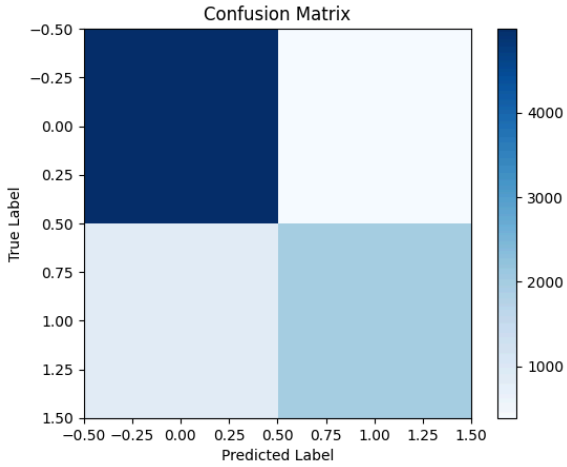



Confusion Matrix
True Label
Predicted Label
−0.50
−0.25
0.00
0.25
0.50
0.75
1.00
1.25
1.50
2250
2000
1750
1500
1250
1000
750
500
250

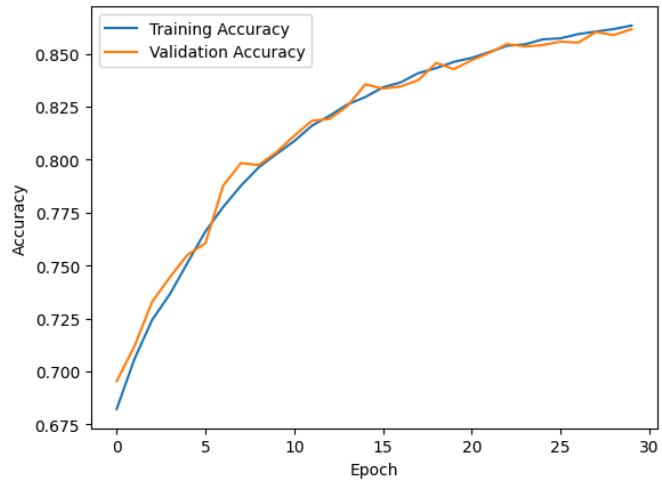
LSTM_30s:

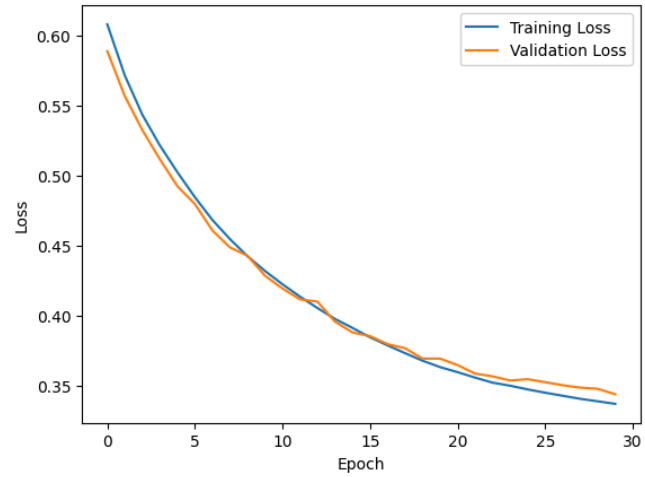

Training Accuracy
Validation Accuracy
Accuracy
Epoch
0.850
0.825
0.800
0.775
0.750
0.725
0.700
0.675
0
5
10
15
20
25
30

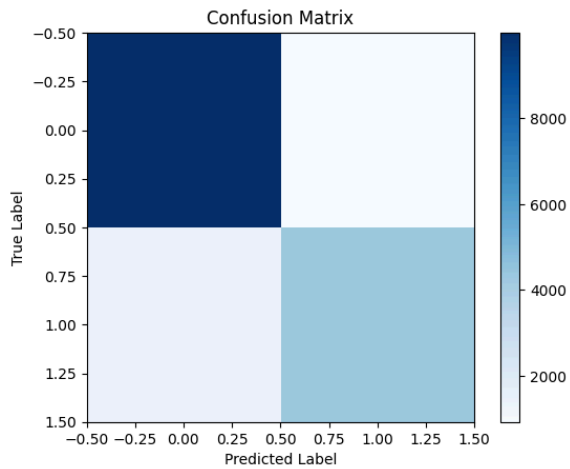

Training Loss
Validation Loss
Loss
Epoch
0.60
0.55
0.50
0.45
0.40
0.35
0
5
10
15
20
25
30


Confusion Matrix
True Label
Predicted Label
−0.50
−0.25
0.00
0.25
0.50
0.75
1.00
1.25
1.50
4000
3000
2000
1000

LSTM_60s:







SVM MODEL:

Data Distribution:

Dataset (Physiological signals data)	**num_train**	**num_test**	**num_items_classes(0,1)**
15s data	31180	7796	25704, 13272
30s data	65702	16426	54162, 27966
60s data	133632	33408	110160, 56880

Model Parameters:

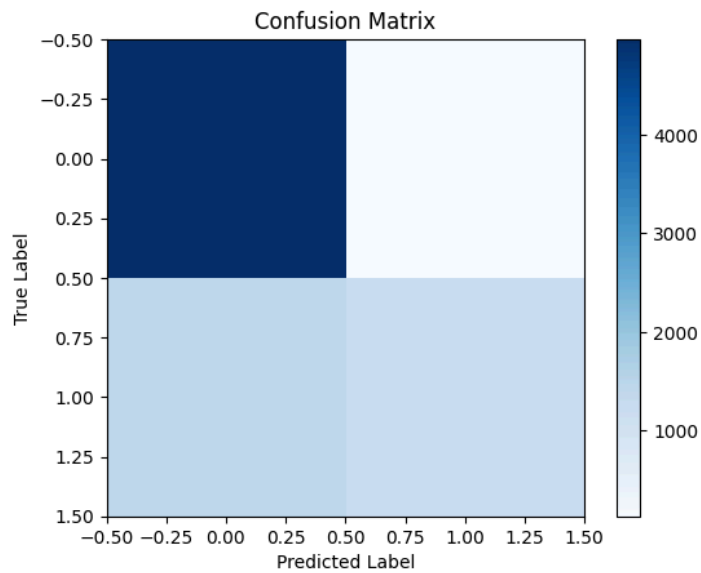
Kernel = 'rbf' (Radial Basis Function): Handles non-linear relationships.
Gamma = 'auto'
Regularization Parameter (C) = 5

Model Performance:

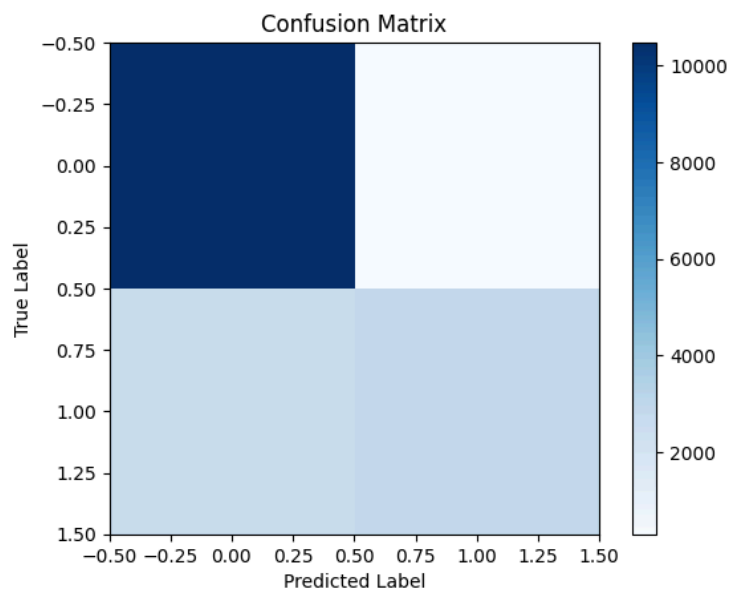
Dataset (Physiological signals data)	Model	Test acc	precision	recall	f1
15s data	SVM	0.79	0.84	0.72/0.79	0.73
30s data	SVM	0.82	0.85	0.75/0.82	0.77
60s data	SVM	0.83	0.85	0.77	0.79

Confusion Matrix:

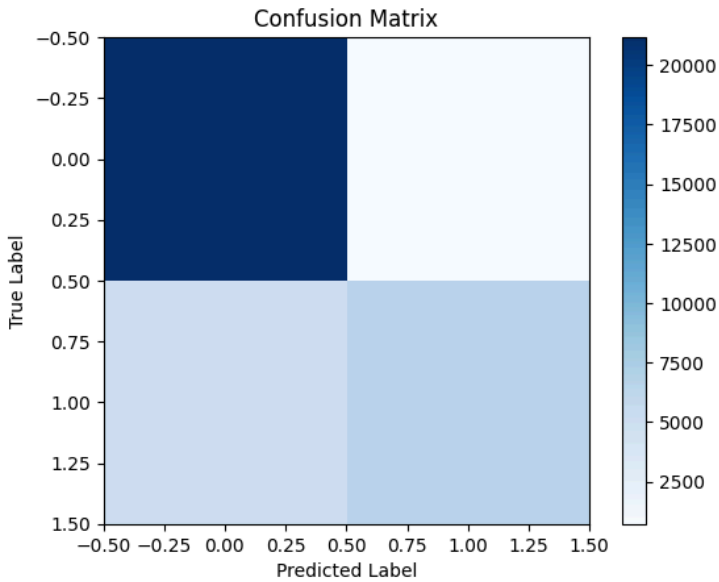
SVM_15s:



SVM_30s:



SVM_60s:



Findings:

- Both LSTM and SVM demonstrated promising results with LSTM having an edge over SVM on test accuracy, precision, recall and f1 score in most cases. Each class is considered equally important, regardless of its size and because of that we have opted for macro average scores.
- It is found that 30s data and 60s data showed improved performance compared to 15s data, suggesting that capturing more extended physiological signal data enhances stress prediction accuracy.
- The confusion matrix shows that both the models performed well for class-0 predictions. However, the model's performance for class-1 predictions is not as good. One of the reasons might be the uneven distribution of data with class-0 having more instances than class-1 instances. Moreover, LSTM showed better performance for class-1 prediction compared to SVM.
- The graph depicts the training and validation accuracy/loss of the LSTM model over 30 epochs. Both training and validation accuracies increase over time. Similarly, both training and validation loss lines are approaching convergence, indicating an effective learning process. The training loss is consistently lower than the validation loss, but both are decreasing at the similar rate, suggesting a good fit without apparent overfitting.

Facial Expressions data:

Data Distribution:

Dataset	num_train	num_validation	num_test	num_item_classes(0,1)
Facial expression	21974	5494	3052	13334,17186

Model Parameters:

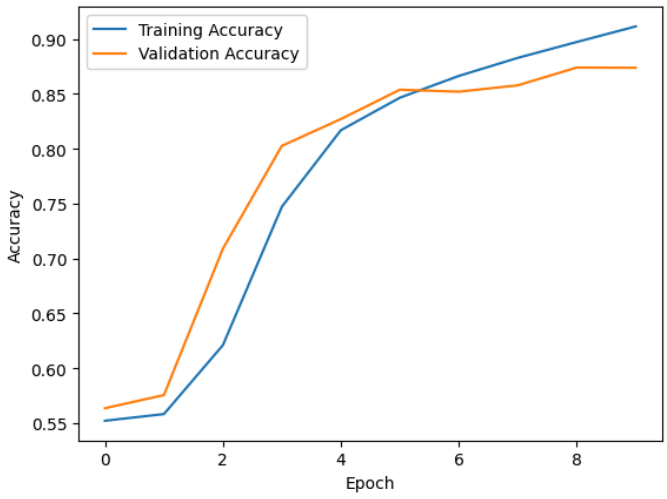
Batch size = 128
Epochs = 10
Optimizer = Adam
Activation = softmax
Loss Function = Categorical Cross Entropy
Lr= 0.001
Input shape: VGG16: (48,48,3); Custom CNN: (48,48,1)

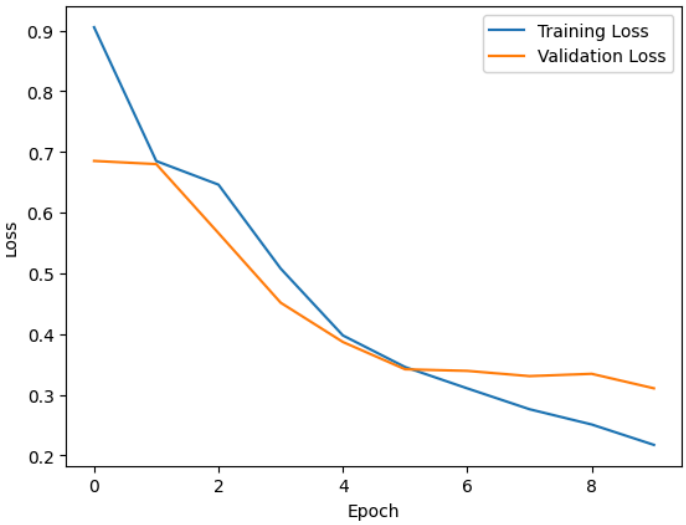
Model Performance:

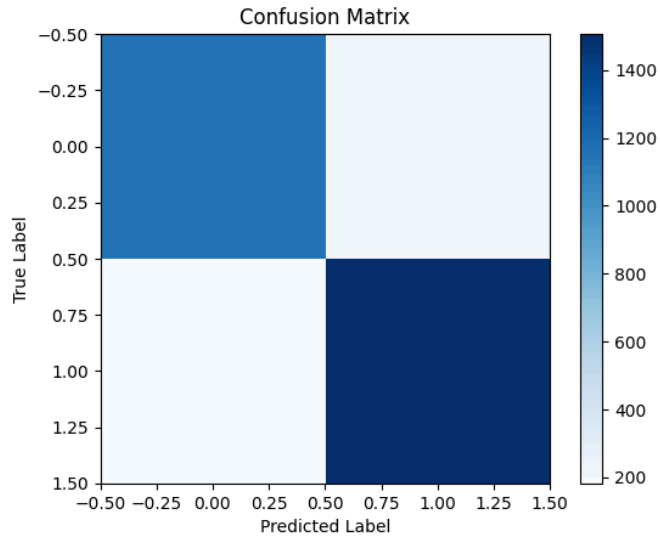
Model	Train acc	Train loss	Val acc	Val loss	Test acc	precision	recall	f1
VGG16	0.9115	0.2174	0.8739	0.3106	0.87	0.87	0.87	0.87
Custom CNN	0.8768	0.2929	0.8085	0.4397	0.81	0.81	0.81	0.81

Graph & Confusion Matrix:

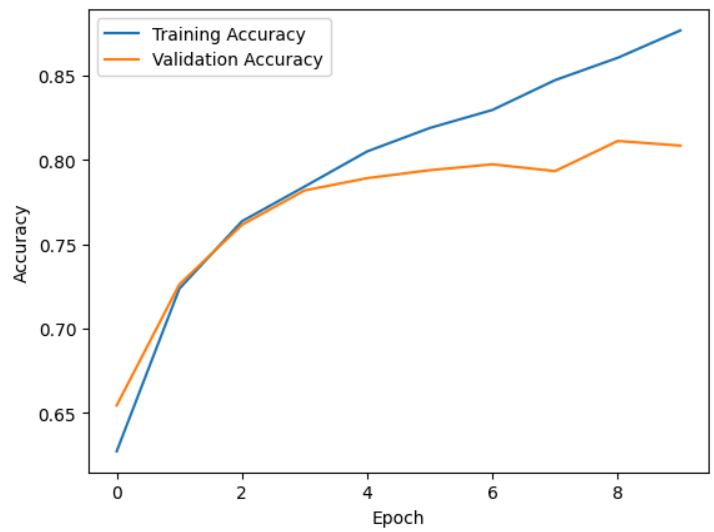
VGG16_2c:

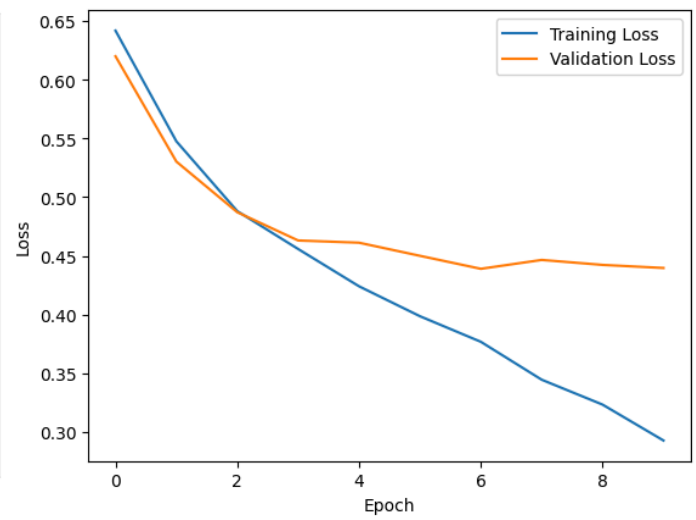


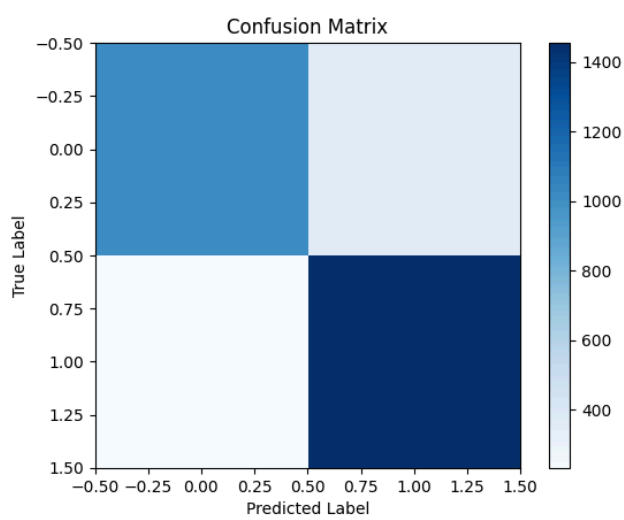




SIMPLE_CNN_2c:







Findings:

- Both VGG16 and Custom CNN achieved high accuracy on the training set, indicating effective learning from the training data. The high accuracy on the validation set suggests that the models generalize well to unseen data.
- VGG16 exhibits a test accuracy of 87%, indicating robust generalization to new and unseen data. Custom CNN achieves a test accuracy of 81%, demonstrating a reasonably good performance on the test set.
- VGG16 outperformed Custom CNN across various metrics, including higher training accuracy, higher validation accuracy, and a slightly better test accuracy. Despite the differences in accuracy, both models demonstrate stable precision, recall, and F1-score values with VGG16 having an edge over our Custom CNN model.
- The confusion matrix shows that both the models performed well for class-0 predictions as well as class-1 predictions, especially class-1 predictions despite having uneven distribution of data. However, the gap is not very big compared to physiological data where the data was more imbalanced.
- From the graph it can be seen that both accuracies are increasing over time, indicating that the model is learning effectively. However, there is a noticeable gap between training and validation accuracy, indicating that the model performs better on the training data than on unseen data. The training loss decreases at a faster rate compared to the validation loss, but both stabilize around after 6-7 epochs.